Problem Statement No.13: Smart Study Generator Agent

The Challenge

Students today face an overwhelming flood of information—lecture notes, textbooks, research papers, YouTube tutorials, and online resources. This content is scattered, unorganized, and often not personalized to their learning pace or style. As a result, students struggle with finding the right study material, summarizing key concepts, and staying exam-ready. Traditional study tools only provide static notes or flashcards, failing to adapt dynamically to student needs. The challenge is to move beyond static resources toward an intelligent, adaptive study companion.

The Objective

Build an agentic AI-powered Smart Study Generator that personalizes and streamlines learning for students. The solution should:

1. **Fuse Learning Resources:** Aggregate content from class notes, textbooks, research articles, and online tutorials into structured, easy-to-digest study material.

2. **Enable Multimodal Input:** Allow students to upload notes, voice queries, or images (e.g., textbook pages), and generate concise summaries, flashcards, or concept maps.

3. **Create a Predictive & Agentic Layer:** Recommend personalized study plans, highlight important topics based on syllabus or past exams, and provide predictive alerts (e.g., "Focus on Topic X—frequently asked in exams").

4. **Visualize Study Progress:** Offer a real-time dashboard showing learning milestones, weak areas, practice quizzes, and performance forecasts.

Tech Stack (Mandatory)

Use **any one** of the following approaches:

- LangFlow + IBM watsonx.ai + Granite Models
- IBM Cloud Agentic Lab with IBM Granite Models
- Replicate with LangChain + RAG + IBM Granite Models